

1 Supplementary Information

2 Origin of crack propagation in LiCoO₂ cathode

3 Wenguang Zhao,^{1,†} Zijian Li,^{2,†} Mingyang Li,^{3,†} Hengyu Ren,¹ Cong Lin,⁴ Haocong
4 Yi,¹ Shengyu Wu,⁵ Xiaohu Wang,¹ Jun Wang,⁶ Bin Fei,² Muyu Zhang,⁷ Mingwei
5 Wang,⁸ Ye Sun,⁸ Guojie Chen,¹ Zhefeng Chen,¹ Weixin Chen,⁹ Xiaoxu Zhao,⁸
6 Shunning Li,^{1,10,*} Dong Zhou,^{11,*} Qinghe Zhao,^{1,5,*} Feng Pan^{1,*}

7

8

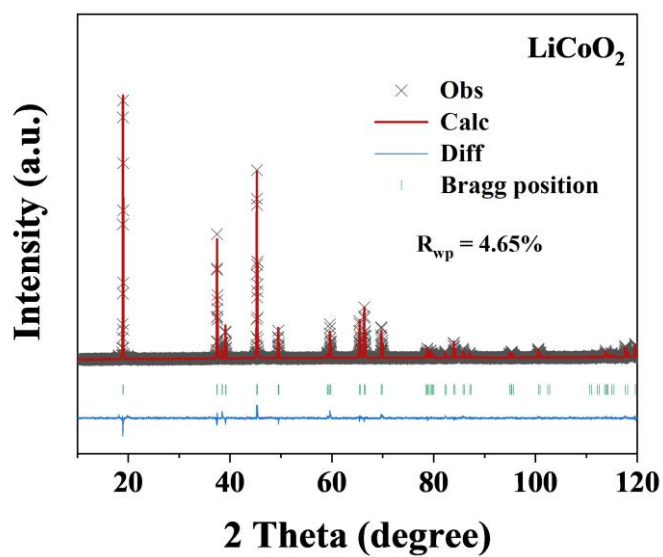


9

10 **Fig. S1.** SEM image of pristine LCO.

11

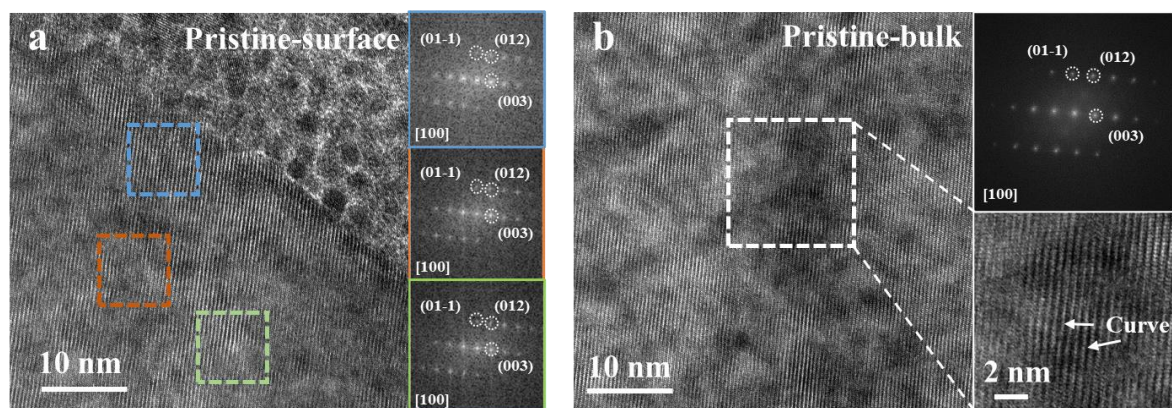
12



13

14 **Fig. S2.** XRD pattern and Rietveld refinement of pristine LCO.

15



17

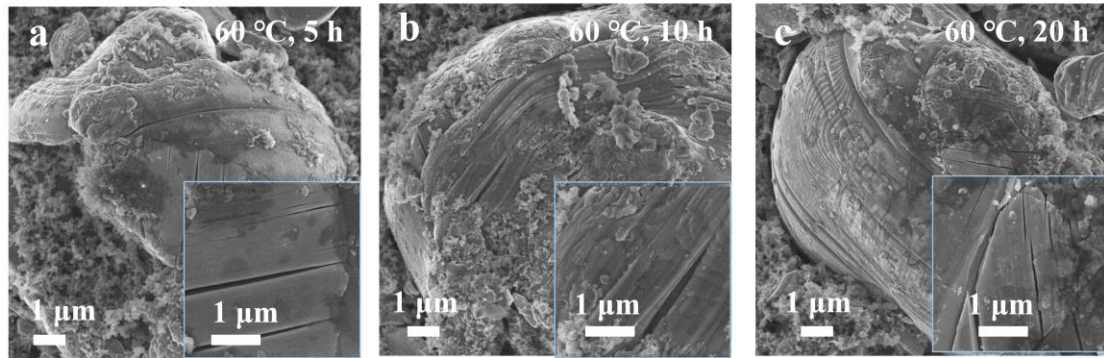
18

19 **Fig. S3.** TEM characterization of pristine LCO.

20 (a) High-resolution TEM image of the surface region, with FFT patterns taken from
21 selected areas marked by blue, orange, and green boxes.

22 (b) High-resolution TEM image of the bulk region, with FFT pattern and corresponding
23 enlarged area marked by white box.

24



26

27 **Fig. S4.** SEM images showing crack evolution of LCO particles after different
28 durations under floating charge of 4.6 V at 60 °C.

29 (a) 5 h.

30 (b) 10 h.

31 (c) 20 h.

32

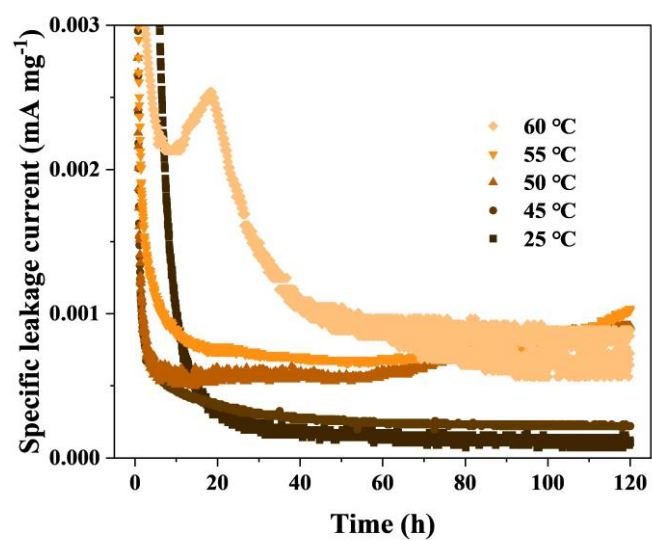


Fig. S5. Current-time plots showing leakage current evolution of LCO||Li cells during float charge at various temperatures.

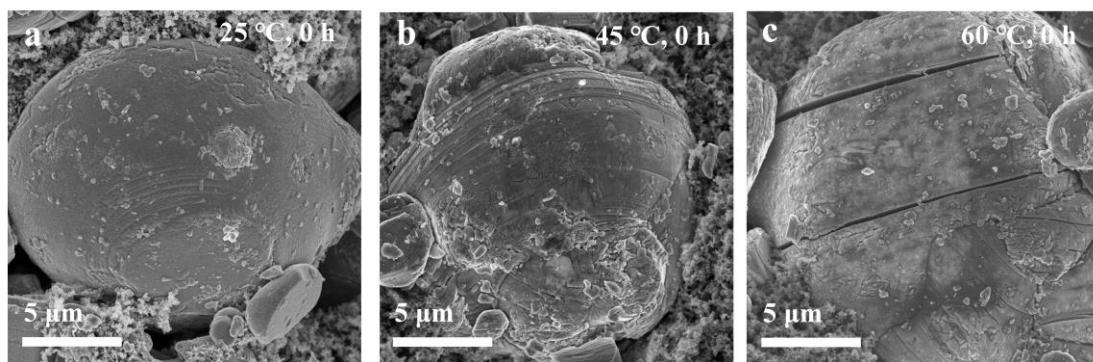


Fig. S6. SEM images of LCO particles prior to float charge (0 h) at different temperatures.

(a) 25 °C.

(b) 45 °C.

(c) 60 °C.

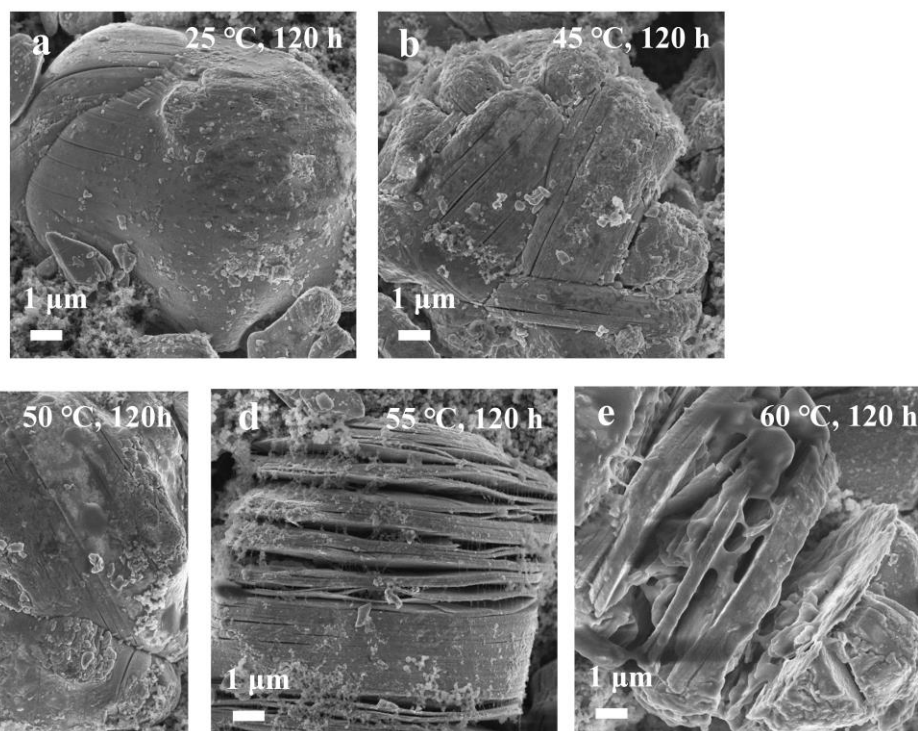


Fig. S7. SEM images of LCO particles after float charge of 120 h at different temperatures.

(a) 25 °C.

(b) 35 °C.

(c) 45 °C.

(d) 50 °C.

(e) 60 °C.

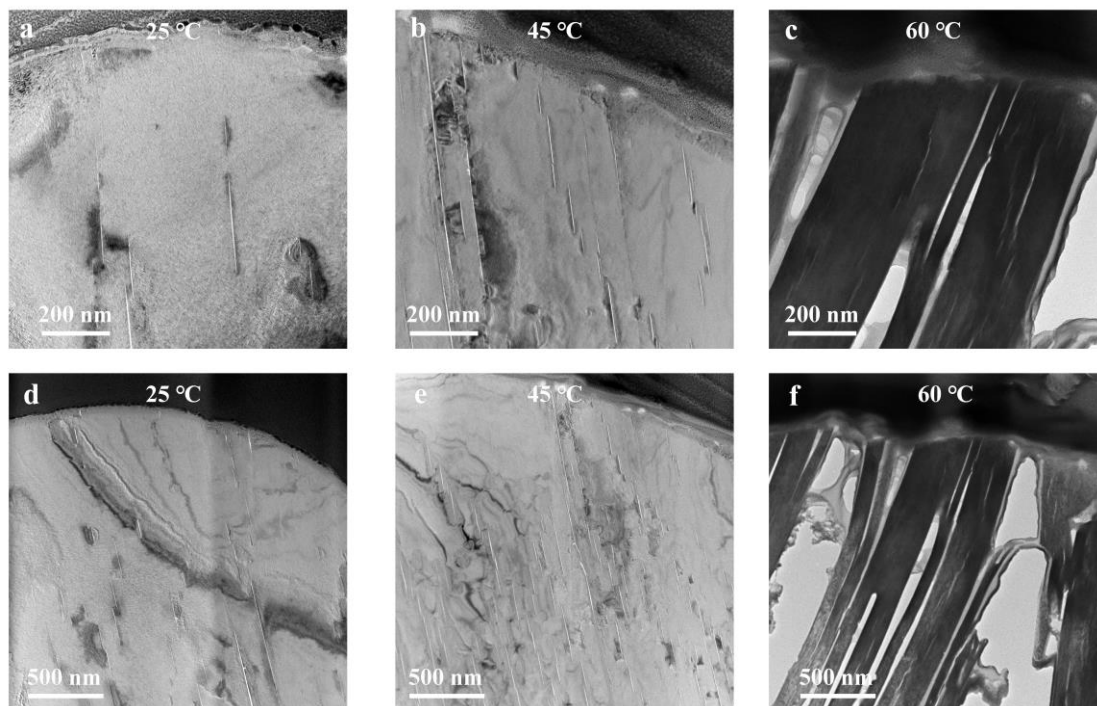


Fig. S8. Cross-sectional TEM and corresponding lower-magnification images of LCO particles after float charge of 120 h at different temperatures.

(a and d) 25 °C.

(b and e) 45 °C.

(c and f) 60 °C.

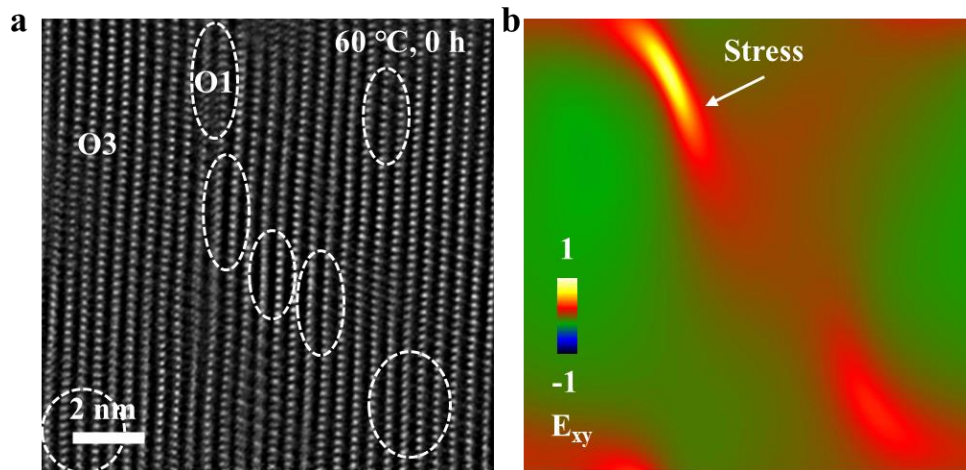


Fig. S9. Microscopic LCO characterizations prior to floating charge (0 h) at 60 °C.

(a) High-resolution STEM image.

(b) Corresponding GPA map.

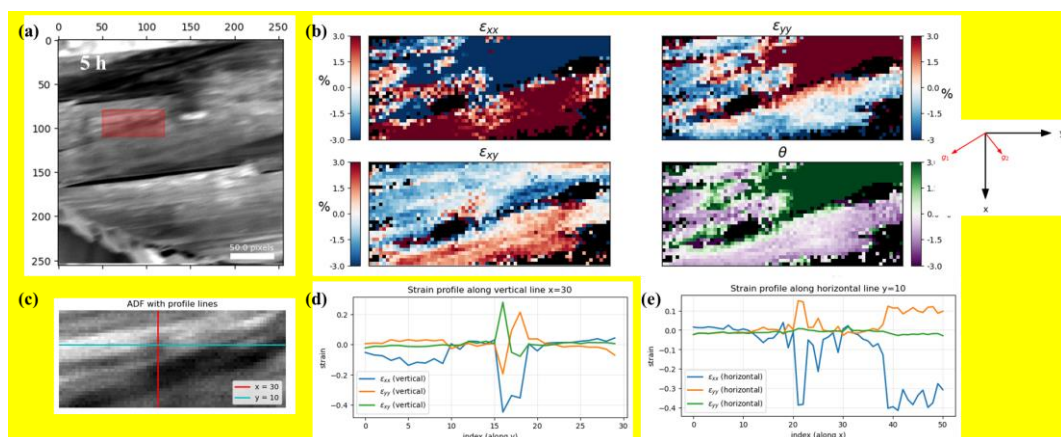


Fig. S10. 4D STEM characterization of LCO after float charging at 60 °C and 4.6 V for 5 h.

(a) STEM image.

(b) Corresponding stress mapping of the selected region.

(c–e) Line-scan profiles of the stress distribution along the x and y directions within the selected area.

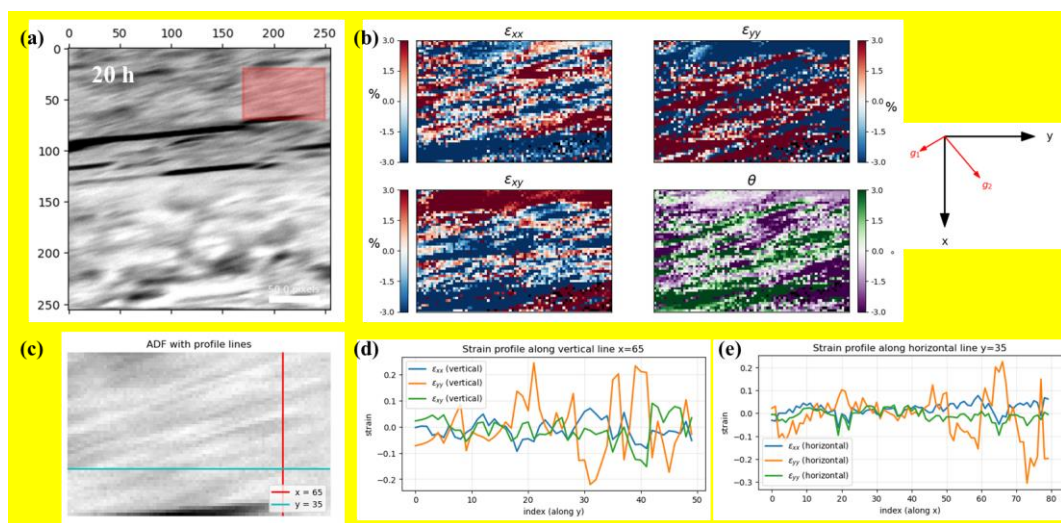


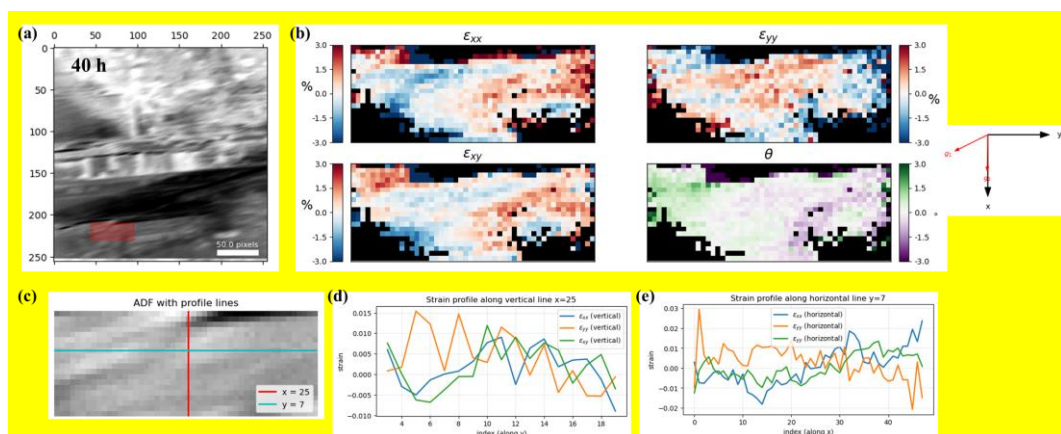
Fig. S11. 4D STEM characterization of LCO after float charging at 60 °C and 4.6 V for

20 h.

(a) STEM image.

(b) Corresponding stress mapping of the selected region.

(c–e) Line-scan profiles of the stress distribution along the x and y directions within the selected area.



85

86 **Fig. S12.** 4D STEM characterization of LCO after float charging at 60 °C and 4.6 V for
 87 40 h.

88 (a) STEM image.

89 (b) Corresponding stress mapping of the selected region.

90 (c–e) Line-scan profiles of the stress distribution along the x and y directions within the
 91 selected area.

92

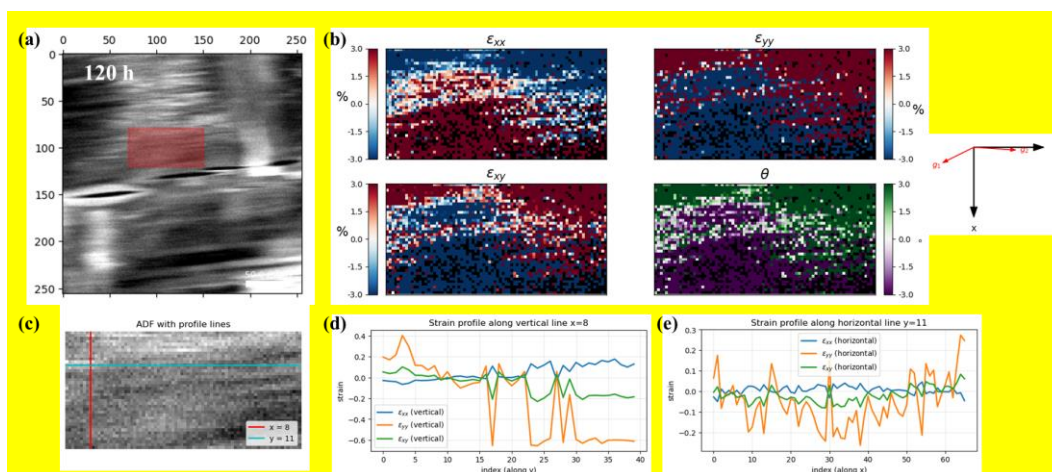


Fig. S13. 4D STEM characterization of LCO after float charging at 60 °C and 4.6 V for

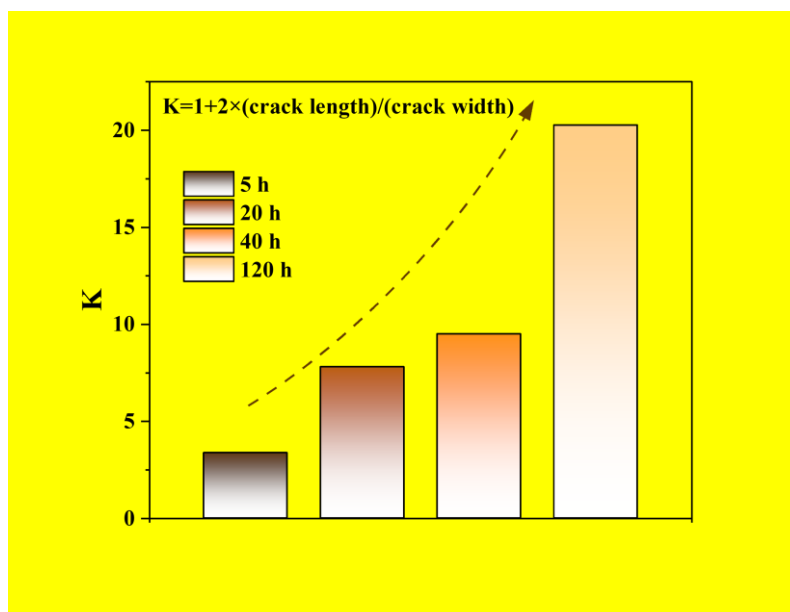
120 h.

(a) STEM image.

(b) Corresponding stress mapping of the selected region.

(c–e) Line-scan profiles of the stress distribution along the x and y directions within the selected area.

102



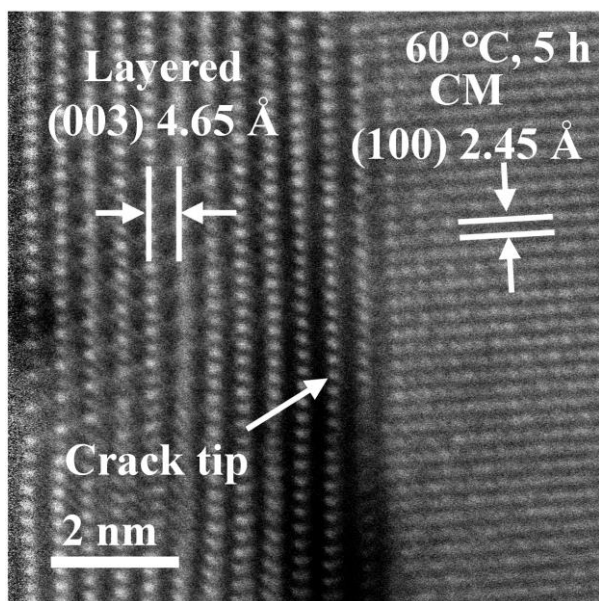
103

104 **Fig. S14.** Stress concentration factor values derived from 4D STEM analysis after float

105 charging at 60°C and 4.6 V for various durations (5, 20, 40, and 120 h).

106

107



108

109 **Fig. S15.** High-resolution STEM image of LCO after float charge for 5 h at 60 °C.

110

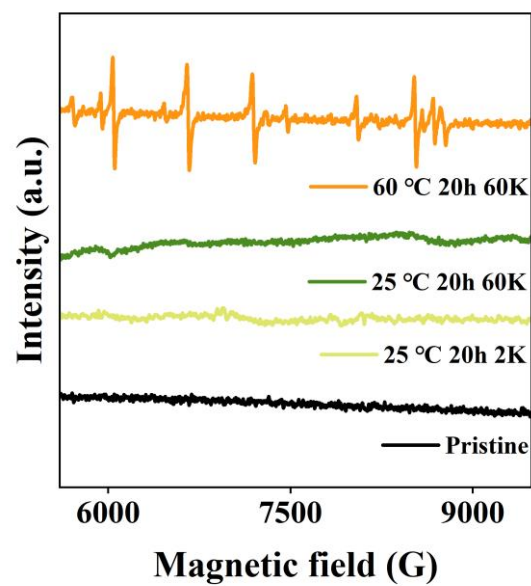
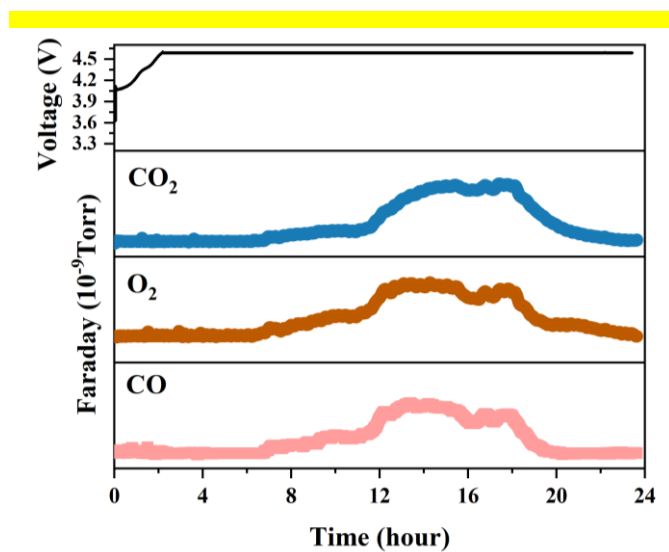


Fig. S16. EPR spectra of the LCO after 20 hours float charge at 25 °C and 60 °C measured at 2K and 60 K.

115



116

117 **Fig. S17.** *In situ* DEMS results, with voltage profile and corresponding CO₂, O₂ and
118 CO signals.

119

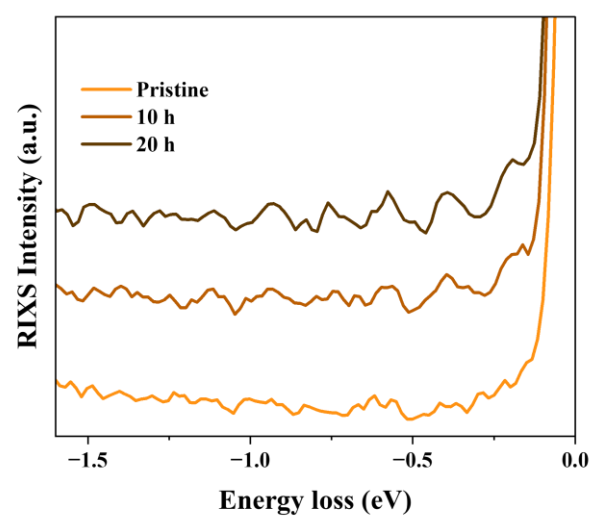
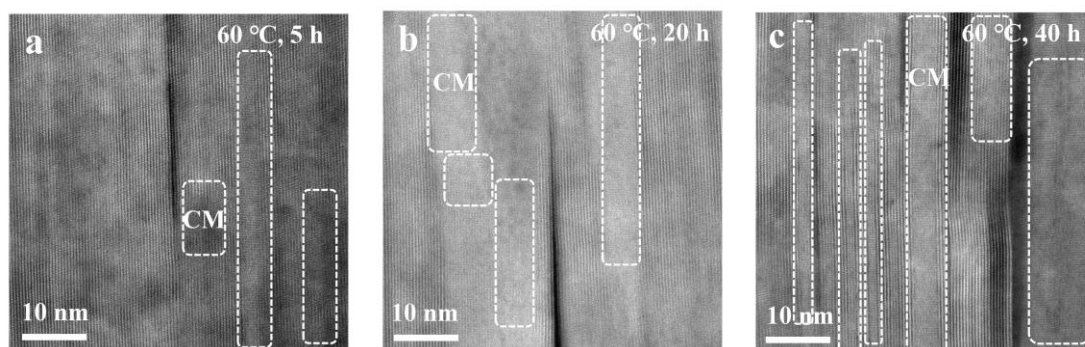


Fig. S18. RIXS spectra of LCO in the pristine state and after float charging at 60 °C and 4.6 V for 10 h and 20 h.

125



126

127 **Fig. S19.** STEM images of LCO particles after float charge under 60 °C for different
128 durations, with the white boxes highlighting the CM phase.

129 (a) 5 h.

130 (b) 20 h.

131 (c) 40 h.

132

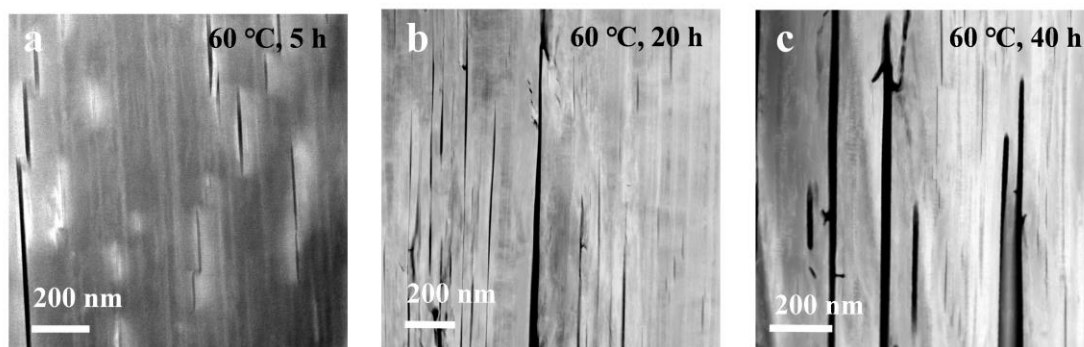


Fig. S20. STEM images showing the crack propagation in LCO after float charge for different durations at 60 °C.

(a) 5 h.

(b) 20 h.

(c) 40 h.

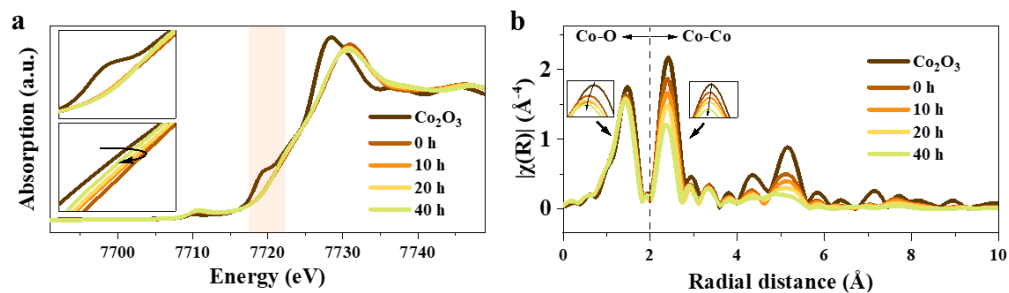
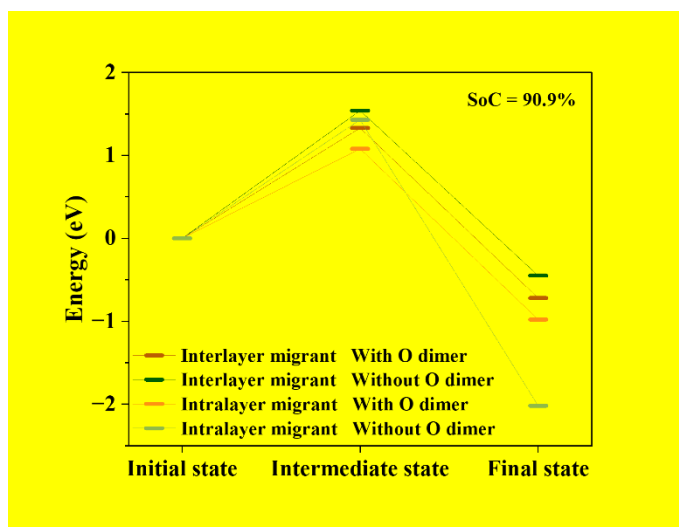


Fig. S21. XANES and EXAFS characterizations of the LCO after float charge for different durations at 60 °C.

(a) Co K-edge XANES spectra, with the inset showing a magnified view at around half-maximum of the absorption edge.

(b) Corresponding Fourier-transformed k₃-weighted EXAFS spectra.

148



149

150 **Fig. S22.** Migration barriers for Co in the presence and absence of O dimers at SoC =
 151 90.9%.

152

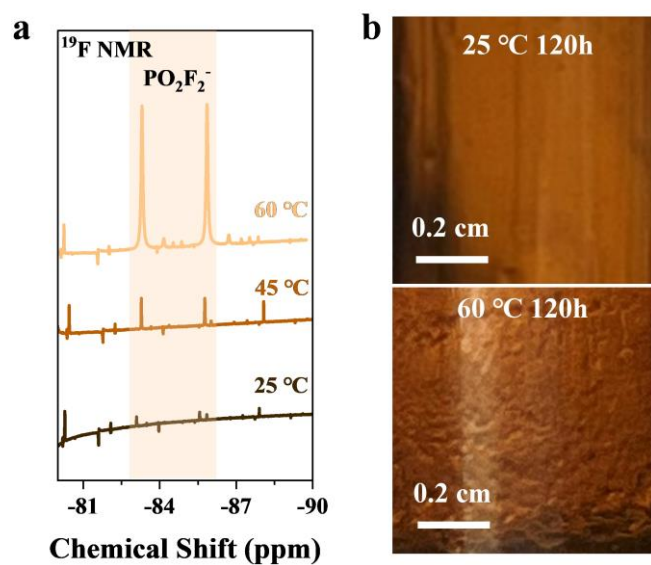


Fig. S23. Electrolyte degradation characterizations

(a) ^{19}F NMR spectra of the electrolytes cycled for 120 h at different temperatures.

(b) Photographs of the inner walls of the glass vials after cycling the electrolytes at

25 °C (top) and 60 °C (bottom) for 120 h.

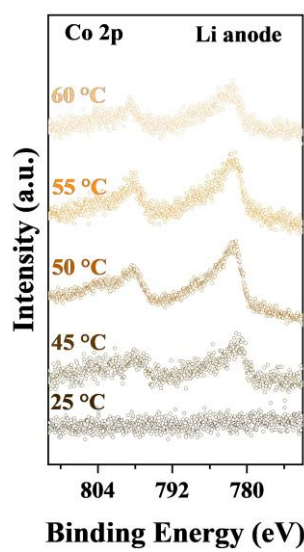


Fig. S24. XPS spectra of Co on Li anodes after float charge at various temperatures.

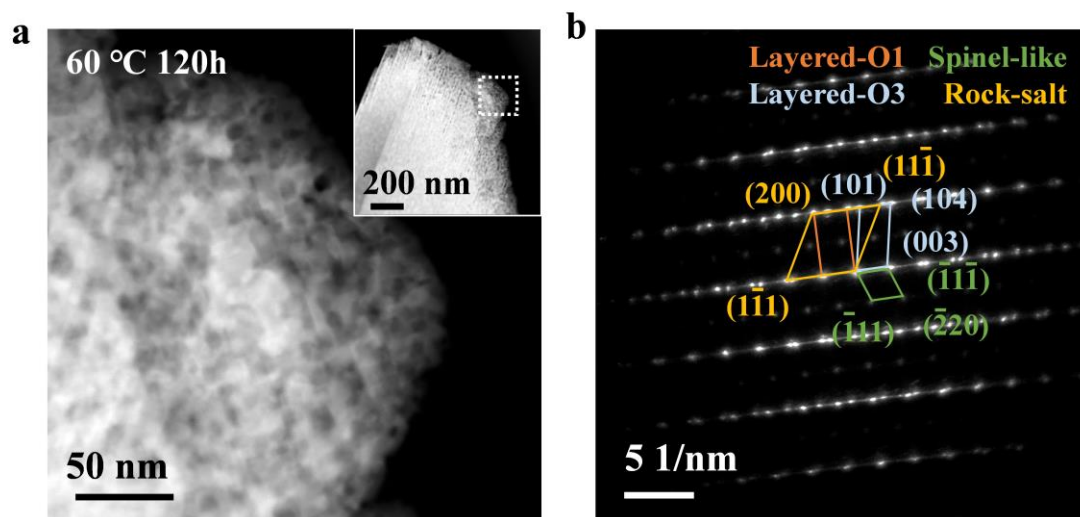


Fig. S25. Microscopic characterizations of the LCO after float charge for 120 h at 60 °C.

(a) STEM image, with the inset showing a corresponding lower-magnification region.

(b) Corresponding SAED pattern, with diffraction spots indexed to layered O3 (blue), layered O1 (orange), spinel (green), and rock-salt (yellow) phases.

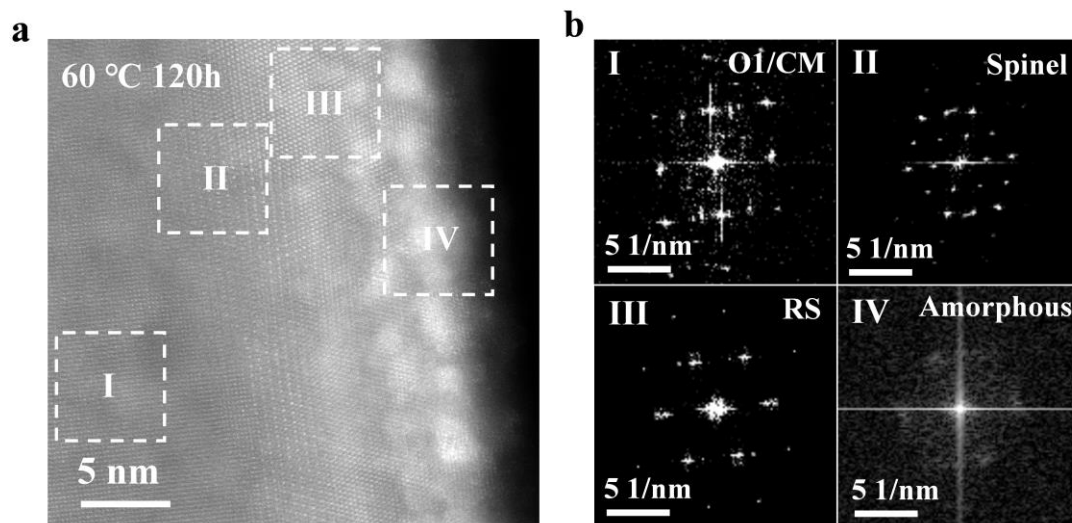


Fig. S26. Atomic-scale characterizations of the LCO after float charge for 120 h at 60 °C. (a) Cross-sectional STEM image. (b) FFT patterns of corresponding regions in (a), assigned to rock-salt, spinel, O1, and the amorphous, respectively.

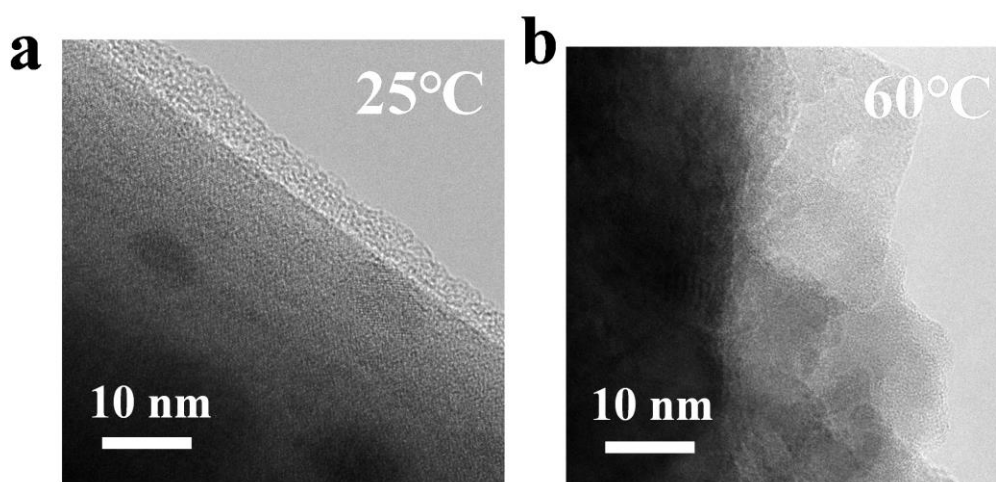


Fig. S27. TEM images of the CEI on LCO after float charge for 120 h at different temperatures.

(a) 25 °C, showing a uniform CEI (~2 nm)

(b) 60 °C, showing a significantly thickened (~15 nm) and porous CEI.

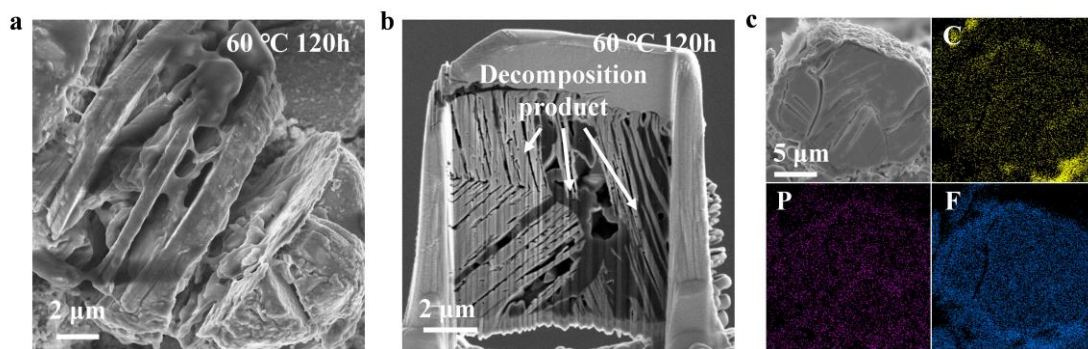


Fig. S28. Microscopic characterizations of the LCO after float charge for 120 h at 60 °C, confirming electrolyte decomposition and penetration.

(a) SEM image.

(b) Cross-sectional SEM image.

(c) EDS elemental mapping.

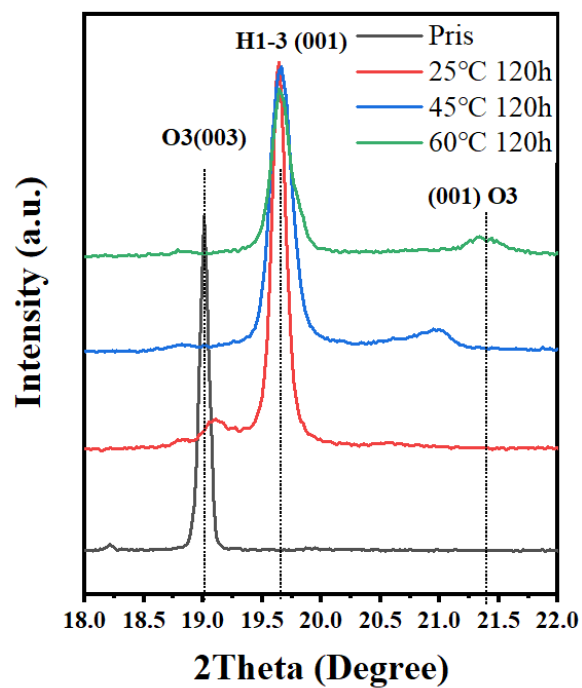


Fig. S29. XRD patterns of the Al-reinforced LCO and after floating for 120 h under various temperatures (25 °C, 45 °C, and 60 °C).

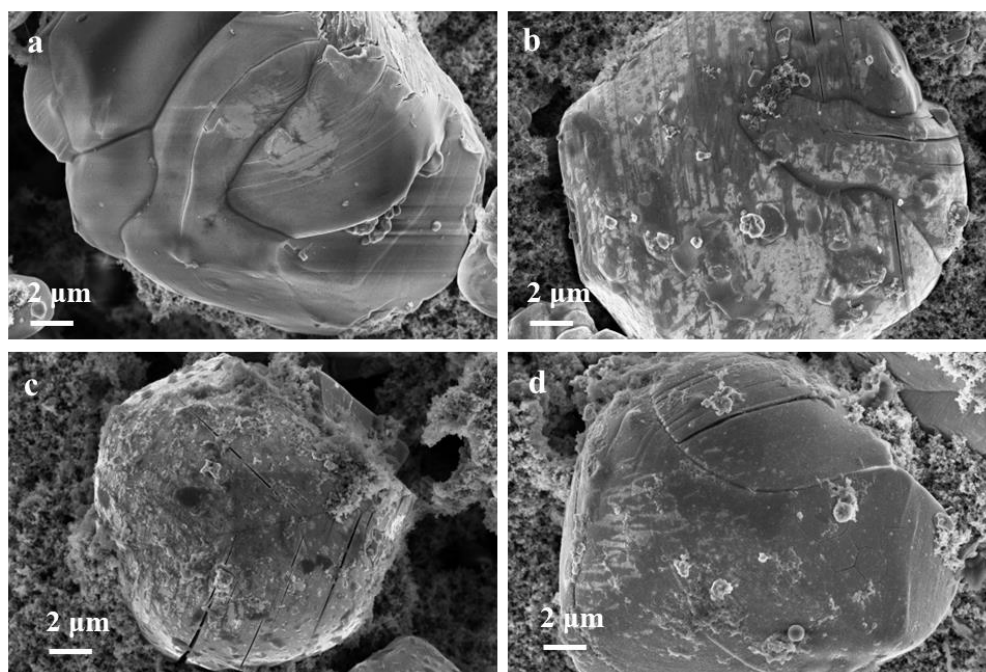


Fig. S30. SEM images of the Al-reinforced LCO particles and after float charge of 120 h at different temperatures.

(a) Pristine particles.

(b) 25 °C.

(c) 45 °C.

(d) 60 °C.

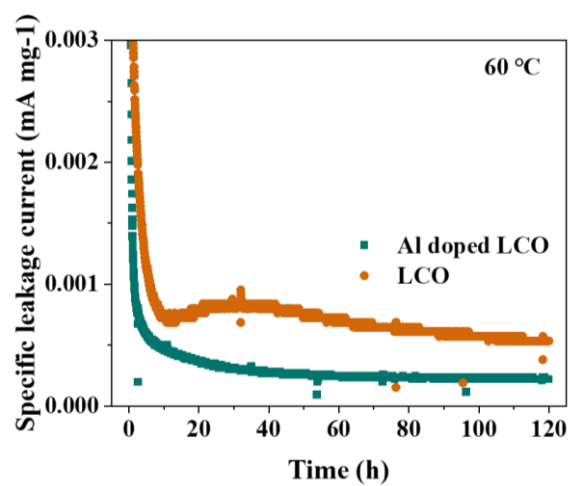


Fig. S31. Comparison of the specific leakage current evolution during 60 °C/4.6 V floating charge between pristine LCO and Al-reinforced LCO.

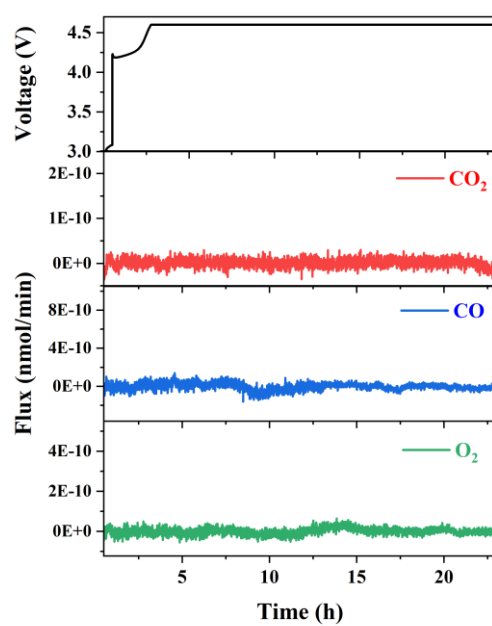


Fig. S32. In situ DEMS of the Al-reinforced LCO during 60 °C/4.6 V floating charge.